Números Complexos

1) definições Gerais

São todos números da forma $z = a + bi, a, b \in \mathbb{R}$

onde $i^2 = -1$.

Temos que:

- a: é a parte Real $\Leftrightarrow \mathrm{Re}(z) = a$
- b: é a parte Imaginária $\Leftrightarrow \mathrm{Im}(z) = b$
- $z \in \mathbb{R} \leftrightarrow \mathrm{Im}(z) = 0$
- z é Imaginário Puro $\leftrightarrow \mathrm{Re}(z) = 0$ e $\mathrm{Im}(z) \neq 0$

2) Igualdade nos Complexos

seja $z = a + bi$; $w = x + yi$

$$z = w \rightarrow \begin{cases} a = x \\ b = y \end{cases}$$

3) Potencias de i

observe:

$$
\begin{array}{llll}
i^1 = i & i^5 = i & \cdots & i^{4k+1} = i \\
i^2 = -1 & i^6 = -1 & \cdots & i^{4k+2} = -1 \\
i^3 = -i & i^7 = -i & \cdots & i^{4k+3} = -i \\
i^4 = 1 & i^8 = 1 & \cdots & i^{4k} = 1
\end{array}
$$

assim $i^{1000} = i^{4 \cdot 250} = 1$

4) Conjugado de um Complexo

Seja $z = a + bi$ seu conjugado ($\bar{z}$ ou z^*)

$$\bar{z} = a - bi$$

assim: $(11 - 3i)^* = 11 + 3i$

Propriedades:

P1) $(z^*)^* = z$

P2) $z \in \mathbb{R} \leftrightarrow z = z^*$

P3) z é Imaginário puro $\leftrightarrow z = -z^*$

$\cdot\, Z \in i\mathbb{R} \leftrightarrow Z = -Z^*$

P4) $Z + Z^* = 2\,\mathrm{Re}(Z)$

P5) $Z - Z^* = 2\,\mathrm{Im}(Z)$

P6) $(Z \pm \omega)^* = Z^* \pm \omega^*$

P7) $(Z \cdot \omega)^* = \overline{Z} \cdot \overline{\omega}$

P8) $(Z^n)^* = (z^*)^n$

P9) $\left(\frac{Z}{\omega}\right)^* = \frac{Z^*}{\omega^*}$

P10) $Z \cdot \overline{Z} = a^2 + b^2$

5) Operações entre Complexos

a) Adição: $(a+bi)+(x+yi) = (a+x) + (b+y)i$

b) diferensa: $(a+bi)-(x+yi) = (a+x) - (b+y)i$

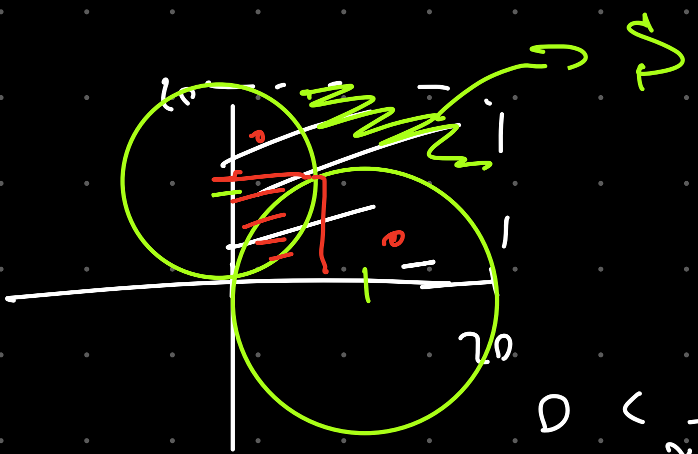
c) Produto: $(a+bi)(x+yi) = ax - yb + (ay + bx)i$

d) Divisão: $\frac{a+bi}{x+yi} = \frac{(a+bi)(x-yi)}{(x+yi)(x-yi)} = \frac{(a+bi)(x-yi)}{x^2+y^2}$

Exemplo: Seja S uma região do plano Cartesiano que consiste em todos os pontos (x,y) tais que as partes Reais e Imaginárias de

$\frac{z}{20}$ e $\frac{20}{\bar{z}}$ estão entre $[0,1)$ $z = x+yi$

$\frac{x+yi}{20}$ e $\frac{20}{x-yi}$ $\quad \frac{x}{20} \in [0,1]$ $\quad \frac{y}{20} \in [0,1)$



$$\frac{20(x+yi)}{x^2+y^2}$$

$$0 < \frac{20}{x^2+y^2}x < 1 \qquad 0 < x < \frac{x^2+y^2}{20}$$

$\square - \square - 2\frac{1}{4} 0$

$$0 < y < \frac{x^2+y^2}{20}$$

$x, y \geq 0$

$x^2+y^2-20x \geq 0$

$x^2+y^2-20x+100 \geq 100$

$(x-10)^2+y^2 \geq 10^2$

e

ai:

$(x^2)+(y-10)^2 \geq 10^2$

$$\sqrt{x+yi} = a+bi$$

$$x+yi = a^2-b^2+2abi$$

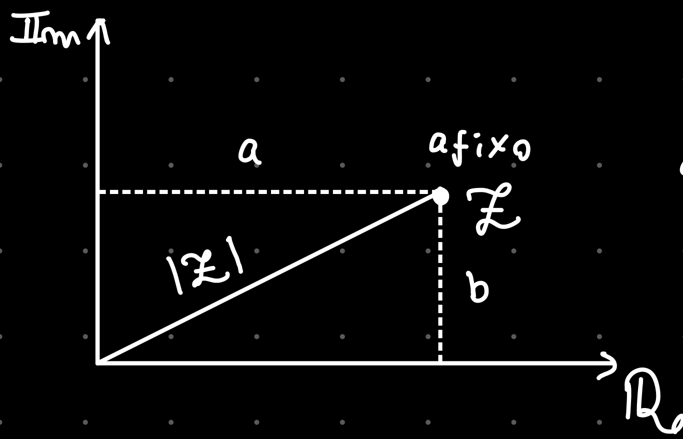
$$i = a^2-b^2-2ab$$

$a = \pm b$

$\begin{cases} 2ab = 1 \end{cases}$ $\quad \frac{\sqrt{2}}{2}$

Módulo de Complexo

Considere a representação de $z = a + bi$ no plano dos complexos



O módulo de z, $|z|$, é a distância do afixo até a origem:

$$|z| = \sqrt{a^2 + b^2}$$

Exemplo: $|3 - 5i| = \sqrt{34}$

Exercício EscNaval Calcule x tal que $|3 - xi + 2i| = 8$

$$\sqrt{9 + (2-x)^2} = 8 \rightarrow \quad (2-x)^2 = 55 \quad 2 - x = \pm\sqrt{55}$$

$$x = 2 \pm \sqrt{55}$$

Propriedades do Módulo

(P1) $|z| = |\bar{z}| = |-z| = |-\bar{z}|$

(P2) $|zw| = |z| \cdot |w|$

(P3) $|z^n| = |z|^n$

(P4) $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$

(P5) $|z|^2 = z^* z$

(P6) $||z|-|w|| \leq |z+w| \leq |z|+|w|$ Desigualdade Triangular

(P7) $\mathrm{Re}(z) \leq |z|$

(P8) $\mathrm{Im}(z) \leq |z|$

Prova De P6

$$|z+w|^2 = (z+w)(z^*+w^*) \rightarrow zz^* + ww^* + zw^* + wz^*$$

$$\rightarrow |z|^2+|w|^2+(zw^*)+(zw^*)^* \rightarrow |z|^2+|w^2|+2\mathrm{Re}(zw^*)$$

$$\rightarrow |z|^2+|w^2|+2|z||w| = |z|^2+|w^2|+2\mathrm{Re}(zw^*)$$

$$(|z|+|w|)^2 \geq |z|^2+|w|^2+2\mathrm{Re}(zw^*)$$

$$(|z|+|w|)^2 \geq (|z+w|)^2$$

$$|z|+|w| \geq |z+w|$$

Ex se z satisfaz $|2z+15| = \sqrt{3}\,|\bar{z}+10|$

Calcule $|z|$

$$|2z+15|^2 = 3\,|\bar{z}+10|^2$$

$$(2z+15)(2\bar{z}+15) = 3(\bar{z}+10)(z+10)$$

$$4z\bar{z}+30z+30\bar{z}+15^2 = 3z\bar{z}+30\bar{z}+30z+300$$

$$z\bar{z} = 300-15^2$$

$$|z|^2 = 75 \qquad |z| = \sqrt{75} = 5\sqrt{3}$$

Se $|w|=3$ $|z|=4$ $|z+w|=6$, calcule $|z-w|$

$|z+w|^2=36 \rightarrow (z+w)(\bar{z}+\bar{w})=36 \rightarrow z\bar{z}+w\bar{w}+z\bar{w}+\bar{z}w=36$

$\rightarrow 25+z\bar{w}+\bar{z}w=36 \rightarrow z\bar{w}+\bar{z}w=11$

- $|z-w|=\varphi \rightarrow |z-w|^2=\varphi^2 \rightarrow (\bar{z}-\bar{w})(z-w)=\varphi^2$

$\rightarrow z\bar{z}+w\bar{w}-z\bar{w}-w\bar{z}=\varphi^2$

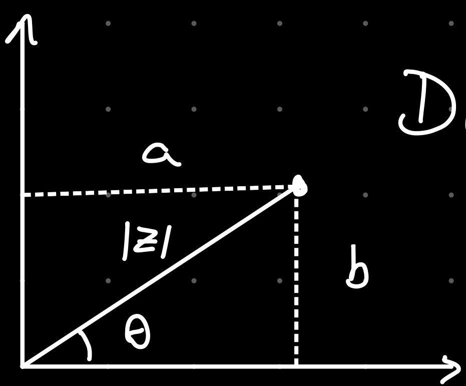
$\rightarrow 25-11=\varphi^2$

$\rightarrow \varphi^2=14$

$\rightarrow \varphi=\sqrt{14}$

Argumento de um Complexo

Considere a representação do $z=a+bi$ no plano de Argand-Gauss



Define-se θ como o argumento de z assim $\arg(z)=\theta$

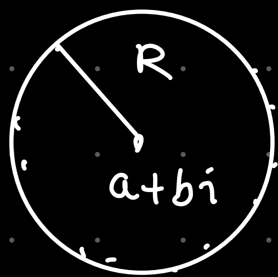
Note que $\operatorname{Tg}\theta=\frac{b}{a}$

assim $\boxed{\operatorname{Tg}(\arg(z))=\frac{\operatorname{Im}(z)}{\operatorname{Re}(z)}}$

Se $z \in \mathbb{R}^*$ $\arg(z)=k\pi$ com $k \in \mathbb{Z}$

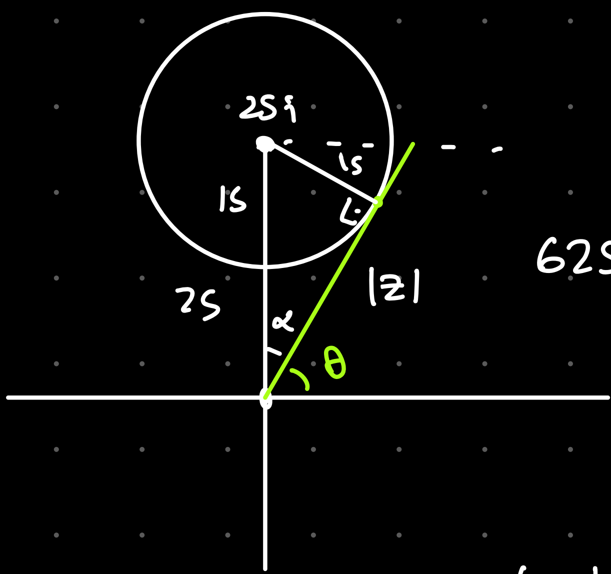
Se $z \in i\mathbb{R}^*$ $\arg(z) = \frac{\pi}{2} + K\pi$ com $K \in \mathbb{Z}$

Importante a Eq $|z-(a+bi)| = R$, $K \in \mathbb{R}^*_+$ representa uma circunferência de centro $(a+bi)$ e raio R.



Z são todos pontos da circunferência

Exercício: Determine o menor argumento que satisfaz $|z-25i| \le 15$



$$625 = 225 + |z|^2$$

$$|z| = 20$$

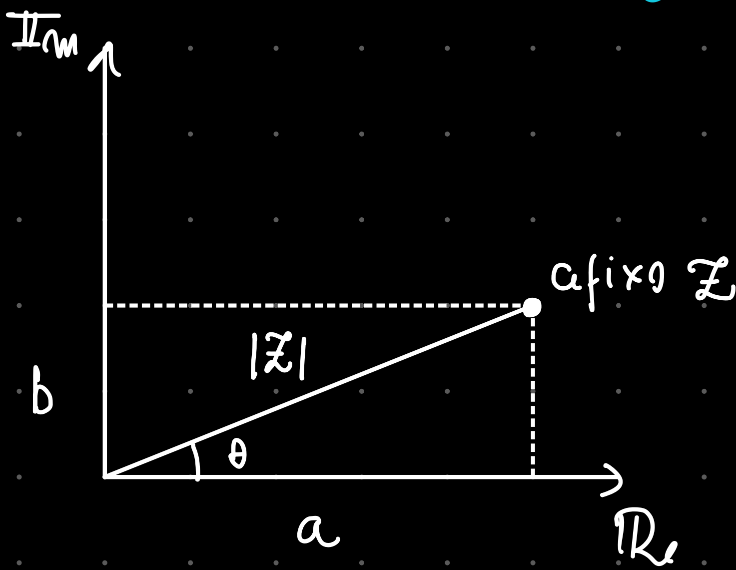
$$\sin\theta = \frac{20}{25}$$

$$\cos\theta = \frac{15}{25}$$

$$20\left(\frac{15}{25} + i\frac{20}{25}\right) = z$$

$$(12 + 16i = z)$$

Forma Trigonométrica



Note que:

$$\cos\theta = \frac{a}{|z|} \quad e \quad \sin\theta = \frac{b}{|z|}$$

$$Re(z) = |z|\cos\theta \quad e \quad Im(z) = |z|\sin\theta$$

$$z = a + bi$$

$$z = |z|\cos\theta + |z|\sin\theta\, i$$

$$z = |z|\underbrace{(\cos\theta + i\sin\theta)}_{cis\theta}$$

assim a forma trigonométrica do complexo é $z = |z|\, cis\, \theta$

Exemplo: $z = 2 - 2\sqrt{3}\, i \rightarrow |z| = 4$

$$z = 4\left(\frac{1}{2} - \frac{\sqrt{3}}{2} i\right) \rightarrow z = 4 \cdot cis\frac{5\pi}{3}$$

Propriedades

seja $z = |z|\, cis\,\alpha$ e $w = |w|\, cis\,\beta$

P1) $z \cdot w = |z||w| \cdot cis(\alpha + \beta)$

Demo:

$$|z|(\cos\alpha + i\sin\alpha)|w|(\cos\beta + i\sin\beta)$$

$|z||w|(\cos\alpha\cos\beta - \sin\alpha\sin\beta + i(\sin\alpha\cos\beta + \cos\alpha\sin\beta))$

$|z||w|(\cos(\alpha+\beta) + i\sin(\alpha+\beta)) = |z||w|\,\text{cis}(\alpha+\beta)$

Geometricamente $\cdot\,\text{cis}(\alpha)$ é rotacionar em α

Em particular $i = \text{cis}\,90°$ então

P2) Divisão

$$\frac{z}{w} = \frac{|z|}{|w|}\,\text{cis}(\alpha-\beta)$$

P3) Conjugado

$$\bar{z} = |z|\,\text{cis}(-\alpha)$$

P4) $\frac{1}{z} = \frac{1}{|z|}\,\text{cis}(-\alpha)$

Igualdade entre $z = |z|\,\text{cis}\,\alpha$ e $w = |w|\,\text{cis}\,\beta$

$$z = w \longleftrightarrow \begin{cases} |z| = |w| \\ \alpha = \beta + 2k\pi \end{cases}$$

Leis De Moivre

(M1) Para $z = |z| \text{cis}\,\theta$ então

$$z^n = |z|^n \text{cis}(n\theta) \quad , n \in \mathbb{Z}$$

Demo: $z \cdot z \cdots z = |z||z| \ldots |z| \text{cis}(\theta + \theta + \ldots + \theta + \theta)$

Exemplo: Calcule $\left(\frac{1-i}{\sqrt{3}+i}\right)^{12} = \frac{\sqrt{2}\,\text{cis}\,\frac{7\pi}{4}}{2\,\text{cis}\,\frac{\pi}{6}} \Rightarrow \left(\frac{\sqrt{2}}{2}\text{cis}\left(\frac{7\pi}{4} - \frac{\pi}{6}\right)\right)^{12}$

$$\Rightarrow \left(\frac{\sqrt{2}}{2}\right)^{12} \text{cis}\left(12\left(\frac{19\pi}{12}\right)\right) = \frac{1}{64}\text{cis}(\pi) = -\frac{1}{64}$$

(M2) Se $w^n = |w| \text{cis}\,\theta$, então

$$w = \sqrt[n]{|w|}\ \text{cis}\left(\frac{\theta + 2k\pi}{n}\right) \quad k = 0, 1, 2, 3, \cdots n$$

Demo:

$w = \rho\,\text{cis}\,\alpha$

$w^n = \rho^n \text{cis}(n\alpha) = |z| \cdot \text{cis}\,\theta$

$|z| = \rho^n \rightarrow \rho = \sqrt[n]{|z|}$ e $\theta = n\alpha$

$\alpha = \frac{\theta}{n}$

IDeiAS Fortes

observe as fatorações

- $\text{cis}\,\theta + 1 = (\cos\theta + 1) + i\sin\theta = \left(2\cos^2\frac{\theta}{2}\right) + i\sin\theta$

$\rightarrow 2\cos^2\theta + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2\cos\frac{\theta}{2}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)$

$$\text{cis}\,\theta + 1 = 2\cos\frac{\theta}{2}\,\text{cis}\,\frac{\theta}{2}$$

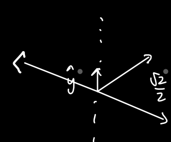
- $\text{cis}\,\theta - 1 = 2i\sin\frac{\theta}{2}\,\text{cis}\,\frac{\theta}{2}$

Exemplo (ITA) $\left((1+\cos 2x) + i\sin 2x\right)^k$

$(1 + \cos 2x + i\sin 2x)^k \rightarrow (1 + \text{cis}\,2x)^k \rightarrow (2\cos x\,\text{cis}\,x)^k$

$\rightarrow 2^k\cos^k x\,\text{cis}\,kx$

Im é $2^k\cos^k\sin(kx)$



$\frac{\sqrt{2}}{2}\hat{x} + \frac{\sqrt{2}}{2}\hat{y}$

$\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\hat{y}$

$\frac{\sqrt{2}}{2}\hat{y}\hat{x}\hat{y} + \frac{\sqrt{2}}{2}\hat{y}\hat{y}\hat{y}$

$-\frac{\sqrt{2}}{2}\hat{x} + \frac{\sqrt{2}}{2}\hat{y}$

$-ava$

Complexos na Trigonometria

Seja $z = \cos\theta + i\sin\theta$ ou $\text{cis}\,\theta$, $|z| = 1$,

Note $\begin{cases} z = \cos\theta + i\sin\theta & (\text{I}) \\ \bar{z} = \cos\theta - i\sin\theta & (\text{II}) \end{cases}$

$(\text{I}) + (\text{II})$: $z + \bar{z} = 2\cos\theta \rightarrow \dfrac{z+\bar{z}}{2} = \cos\theta$

$(\text{I}) - (\text{II})$: $z - \bar{z} = 2i\sin\theta \rightarrow \dfrac{z-\bar{z}}{2i} = \sin\theta$

Como $|z| = 1$ então $z\bar{z} = 1$ então $\bar{z} = \dfrac{1}{z}$

$\cos\theta = \dfrac{z + \frac{1}{z}}{2}$ $\quad \sin\theta = \dfrac{z - \frac{1}{z}}{2i}$ $\quad$ Para $z = \text{cis}\,\theta$

Em especial $z^n = \cos(n\theta) + i\sin(n\theta)$

então

$$\cos(n\theta) = \frac{z^n + \frac{1}{z^n}}{2} \quad \text{e} \quad \sin(n\theta) = \frac{z^n - \frac{1}{z^n}}{2i}$$

Por exemplo se $z = \text{cis}\frac{\pi}{5}$ $\quad \cos\frac{\pi}{5} = \,?$

$\cos\frac{\pi}{5} = \dfrac{z + \frac{1}{z}}{2}$ $\sin\frac{4\pi}{5} = \dfrac{z^4 - \frac{1}{z^4}}{2i}$

Exercício (ITA 2018) Calcule

$S = \cos\frac{\pi}{7} + \cos\frac{3\pi}{7} + \cos\frac{5\pi}{7}$ Seja $z = \operatorname{cis}\frac{\pi}{7}$

$$S = \frac{z + \frac{1}{z}}{2} + \frac{z^3 + \frac{1}{z^3}}{2} + \frac{z^5 + \frac{1}{z^5}}{2}$$

$$2S = z + \frac{1}{z} + z^3 + \frac{1}{z^3} + z^5 + \frac{1}{z^5} \quad (\cdot z^5$$

$$2S \cdot z^5 = z^6 + z^4 + z^8 + z^2 + z^{10} + 1$$

$$2S z^5 = z^{10} + z^8 + z^6 + z^4 + z^2 + 1$$

Se $z = \operatorname{cis}\frac{\pi}{7} \rightarrow z^7 = -1$

$$\rightarrow 2z^5 S = z^7 z^3 + z^7 z + z^6 + z^4 + z^2 + 1$$

$$2z^5 S = z^6 + z^4 - z^3 + z^2 - z + 1$$

$$(2S-1)z^5 = \underbrace{z^6 - z^5 + z^4 - z^3 + z^3 - z + 1}_{PG_2}$$

$$(2S-1)z^5 = \frac{((-z)^7 - 1)}{} \rightarrow \left(\frac{\cancel{1-1}}{-z-1}\right)^{0} = (2S-1)z^5$$

$-z-1$

$2S = 1 \rightarrow S = \frac{1}{2}$

Exemplo$_2$ (IME)

Seja $\frac{a}{b} = \sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{\pi}{5}$ determine $b \in \mathbb{Z}^*$

Seja $z = \text{cis}\frac{\pi}{14}$

$$\left(\frac{z-\frac{1}{z}}{2i}\right)\left(\frac{z^3-\frac{1}{z^3}}{2i}\right)\left(\frac{z^5-\frac{1}{z^5}}{2i}\right) = \frac{1}{b}$$

$$\rightarrow (z^2-1)(z^6-1)(z^{10}-1) = \frac{-8i}{b}z^9$$

$$\rightarrow (z^8-z^2-z^6+1)(z^{10}-1) = -\frac{8i}{b}z^9$$

$$(z^{18}-z^{12}-z^{16}+z^{10}-z^8+z^2+z^6-1) = -\frac{8i}{b}z^9$$

$$(-z^4-z^{12}+z^2+z^{10}-z^8+z^2+z^6-1)$$

$$(\underbrace{-1+z^2-z^4+z^6-z^8+z^{10}-z^{12}}_{SPG}) = -\frac{8i}{b}z^9 - z^2$$

$$-\frac{8i}{b}z^9 - z^2 = \frac{(-1)\cdot[(-z^2)^7-1]}{-z^2-1} \sim 0$$

$-8iz^9 \quad z^2$

$\frac{\ldots}{n} = z$

$-\frac{8i}{b} = z^{-7} \rightarrow \text{cis}\frac{-\pi}{2} \qquad -i$

$-\frac{8i}{b} = -i \qquad b = \frac{1}{8}$

Fazer as 5 questões seguida.

Exercício 4 resolva $\sin x + \sin 2x + \sin 3x = \cos x + \cos 2x + \cos 3x$

Seja $z = \text{cis}\, x$

$$\left(\frac{z-\frac{1}{z}}{2i}\right) + \left(\frac{z^2-\frac{1}{z^2}}{2i}\right) + \left(\frac{z^3-\frac{1}{z^3}}{2i}\right) = \frac{z+\frac{1}{z}}{2} + \frac{z^2+\frac{1}{z^2}}{2} + \frac{z^3+\frac{1}{z^3}}{2}$$

$$\frac{z^4-z^2+z^5-z+z^6-1}{2i} = \frac{z^4+z^2+z^5+z+z^6+1}{2}$$

$$z^4-z^2+z^5-z+z^6-1 = i(z^4+z^2+z^5+z^6+z+1)$$

$$-1-z-z^2+z^4+z^5+z^6 = i(1+z+z^2+z^4+z^5+z^6)$$

$$-(1+z+z^2)+(z^4+z^5+z^6) = i(1+z+z^2+z^3+z^4+z^5+z^6-z^3)$$

$$-(1+z+z^2)+z^4(1+z+z^2) = i(1+z+z^2+z^3+z^4+z^5+z^6-z^3)$$

$$(1+z+z^2)(z^4-1) = i(1+z+z^2+z^4+z^5+z^6)$$

$$i(1+z+z^2+z^4(1+z+z^2))$$

$$(1+z+z^2)(z^4-1) = i((1+z+z^2)(z^4+1))$$

Para $z^2+z+1=0 \rightsquigarrow z = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$ $240°$, $120°$

Otherwise $z^4 - 1 = i z^4 + i$

$$z^4(1-i) = (i+1)$$

$$z^4 = \frac{1+i}{1-i} \to \quad \to x = \frac{\pi}{8} + \frac{k\pi}{2}$$

Forma Polar de Complexos

é possível provar a identidade:

$$e^{i\theta} = \text{cis}\,\theta \qquad e^{i\pi} = \text{cis}\,\pi$$

$$\to e^{i\pi} = -1$$

$$\to e^{i\pi} + 1 = 0$$

$$e^{i\theta} = \sum \frac{(i\theta)^n}{n!}$$

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2} - \frac{i\theta^3}{6} + \frac{\theta^4}{24} + \frac{i\theta^5}{120} \cdots$$

$$e^{i\theta} = \Big(\underbrace{1 - \frac{\theta^2}{2} + \frac{\theta^4}{24} - \frac{\theta^6}{6!} \cdots}_{\cos\theta}\Big) + i\Big(\underbrace{\theta - \frac{\theta^3}{6} + \frac{\theta^5}{120} - \frac{\theta^7}{7!} \cdots}_{\sin\theta}\Big)$$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{i\theta} = \text{cis}\,\theta$$

Exercício $(x+k)^n - x^n - k^n = 0$ e $x = k e^{\frac{2\pi i}{3}}$

$$\left(\cancel{k}\left(1+e^{\frac{2\pi i}{3}}\right)\right)^n = \cancel{k^n}\left(e^{\frac{2\pi n i}{3}}+1\right)$$

$$\left(1+\text{cis}\frac{2\pi}{3}\right)^n = \text{cis}\frac{2\pi n}{3}+1$$

$$\left(\cancel{2}\cos\frac{\pi}{3}\,\text{cis}\frac{\pi}{3}\right)^n = 2\cos\frac{n\pi}{3}\,\text{cis}\frac{n\pi}{3}$$

$\hookrightarrow \frac{1}{\cancel{2}}$

$$\text{cis}\frac{n\pi}{3} = 2\cos\frac{n\pi}{3}\,\text{cis}\frac{n\pi}{3}$$

$$2\cos\frac{n\pi}{3} = 1 \rightarrow \cos\frac{n\pi}{3} = \frac{1}{2} \rightarrow \frac{n\pi}{3} = \pm\frac{\pi}{3} + 2k\pi$$

$n = 6k \pm 1$ com $k \in \mathbb{Z}_+^*$ ou $n = 1$

Raízes da Unidade

Seja $n \in \mathbb{Z}_+$, $n \geq 2$. As raízes da equação $z^n = 1$ são chamadas de raízes n-ésimas da unidade

Observe: $1 = \text{cis}\,0$, $z^n = \text{cis}\,0$

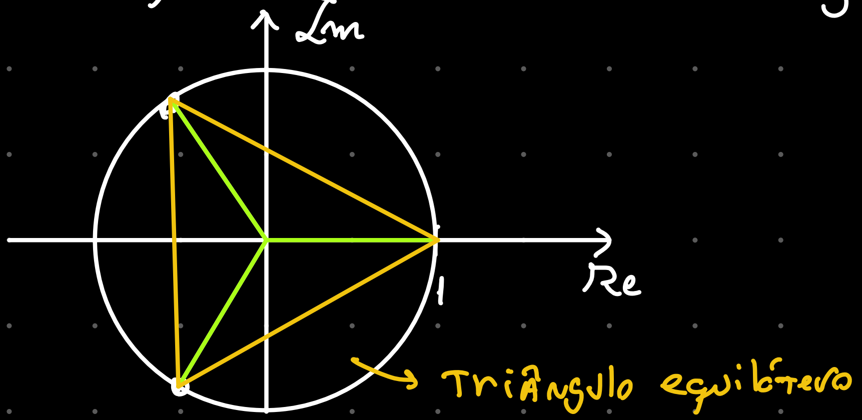
$$z = \sqrt[n]{1}\ \text{cis}\left(\frac{0+2k\pi}{n}\right) \qquad k \in \{0, 1 \dots n-1\}$$

assim $Z_0 = 1$, $Z_1 = \text{cis}\,\frac{2\pi}{n}$, $Z_2 = \text{cis}\,\frac{4\pi}{n}$, $Z_3 = \text{cis}\,\frac{6\pi}{n}$

$\ldots\ Z_{n-1} = \text{cis}\left(\frac{2(n-1)\pi}{n}\right)$

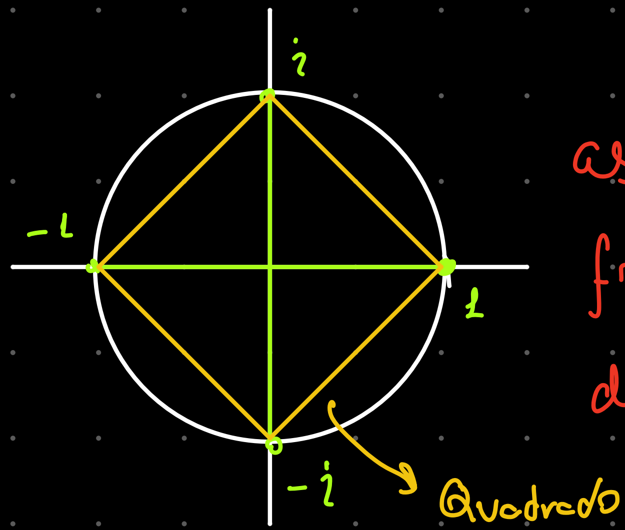
Suponhotemos $n=3$

$Z_0 = 1$, $Z_2 = \text{cis}\,\frac{2\pi}{3}$, $Z_3 = \text{cis}\,\frac{4\pi}{3}$



Suponhotemos $n=4$

$Z_0 = 1$, $Z_2 = \text{cis}\,\frac{\pi}{2}$, $Z_3 = \text{cis}\,\pi$, $Z_4 = \text{cis}\,\frac{3\pi}{2}$



as Raízes n-ésimas formam polígonos regulares de n lados, centro em $(0,0)$ e raio $= 1$ → para raiz da unidade

Seja $Z_1 = \omega$, $Z_2 = \omega^2$, $Z_3 = \omega^3 \cdots Z_{n-1} = \omega^{n-1}$

$1, \omega, \omega^2, \omega^3, \omega^4, \omega^5 \ldots, \omega^{n-1}$

e sua soma é sempre 0

$$S_n = \frac{\omega^n - 1}{\omega - 1} \Rightarrow \omega^n = 1 \rightarrow S_n = 0$$

Exemplo (IME) seja ω raiz cúbica da unidade $\omega \neq 1$ Calcule $(1-\omega)^6$

$1, \omega, \omega^2 : \omega^3 = 1$

$(3\omega(\omega-1))^2$

$9\omega^2(\omega-1)^2$

$9(\omega^4 - 2\omega^3 + \omega^2)$

$\omega \quad 1$

$9(\omega^2 + \omega + 1 - 3)$

$9(-3) \Rightarrow -27$

$(1-\omega)^6$

$(1 - \text{cis}\frac{2\pi}{3})^6$

$(-2i\,\text{sin}\frac{\pi}{3}\,\text{cis}\frac{\pi}{3})^6$

$\cancel{-2^6}\, i^6 \left(\frac{\sqrt{3}}{\cancel{2}}\right)^6 \underbrace{\text{cis}\frac{6\pi}{3}}_{1}$

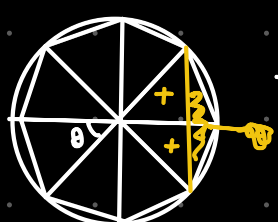
$-1(\sqrt{3})^6$

$-1((\sqrt{3})^2)^3$

-3^3

-27

Exercício ITA 2017 Área do polígono cujos vértices são as raízes de $1+z+z^2+\ldots+z^7=0$



$360 \,|\, 8$
$32 \quad 45$
40

z é raiz octa da unidade

1 não faz parte

$\frac{1}{2} \cdot \frac{\sqrt{2}}{2} \cdot 8 \rightarrow \frac{4\sqrt{2}}{2} \rightarrow 2\sqrt{2} -$ [illegible]

Exercício 2 Simulado Nacional Determine z

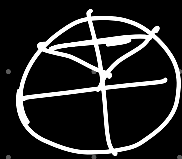
tal que $|z|=1$, $|z^3-\bar{z}^3|=\sqrt{2}$ e $\frac{\pi}{2} \leq \arg(z^3) \leq \pi$

$\downarrow$

$\text{cis}\,\theta$

$$|\text{cis}\,3\theta - \text{cis}\,{-3\theta}| = \sqrt{2}$$

$$\cancel{\cos 3\theta} + i\sin 3\theta$$
$$-\cancel{\cos 3\theta} + i\sin 3\theta$$



$$|2i\sin 3\theta| = \sqrt{2}$$

$$2\cdot\sin 3\theta = \sqrt{2}$$

$$3\theta = \frac{\pi}{4} + 2k\pi$$

$$3\theta = \pi - \frac{\pi}{4} + 2k\pi$$

$$\frac{3\pi}{4} + 2k\pi$$

$$\begin{cases} \frac{\pi}{12} + \frac{2k\pi}{3} \\ \frac{\pi}{4} + \frac{2k\pi}{3} \end{cases}$$

$$\frac{\pi}{6} \leq \theta \leq \frac{\pi}{3}$$

$$\frac{\pi}{4} = z$$

$$\text{cis}\,\frac{\pi}{4}$$

$$z = \frac{\sqrt{2}}{2} + \frac{i\sqrt{2}}{2}$$

Exercício 3 Determine a Σ dos pontos reais das soluções de $(z+1)^5 + z^5 = 0$

$$(z+1)^5 = -z^5$$

$$\frac{(z+1)^5}{z^5} = -1 \rightarrow \left(\frac{z}{z} + \frac{1}{z}\right)^5 = -1$$

$$\left(1 + \frac{1}{z}\right)^5 = -1$$

$$1 + \frac{1}{z} = \sqrt[5]{\operatorname{cis}\pi} \Rightarrow \operatorname{cis}\left(\frac{\pi + 2k\pi}{5}\right) = 1 + \frac{1}{z}$$

$$\operatorname{cis}\theta_k - 1 = \frac{1}{z_k} \rightarrow$$

$$z_k = \frac{1}{\operatorname{cis}\theta_k - 1}$$

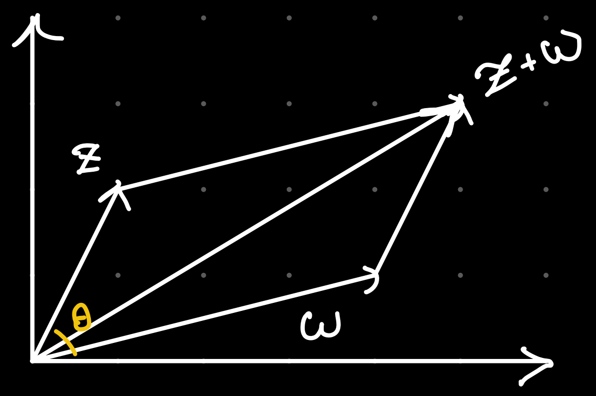
$$z_k = \frac{1}{2i\sin\frac{\theta_k}{2}\operatorname{cis}\frac{\theta_k}{2}} \rightarrow z_k = \frac{-i\sin\frac{\theta_k}{2}\operatorname{cis}\left(-\frac{\theta_k}{2}\right)}{2\sin\frac{\theta_k}{2}}$$

$$z_k = \frac{-\operatorname{cis}\frac{\pi}{2}\operatorname{cis}\left(-\frac{\theta_k}{2}\right)\sin\frac{\theta_k}{2}}{2\sin^2\frac{\theta_k}{2}}$$

$$z_k = \frac{-\operatorname{cis}\left(\frac{\pi}{2} - \frac{\theta_k}{2}\right)}{2\sin\frac{\theta_k}{2}} \rightarrow$$

Soma de Vetores

Observe $z, \omega \in \mathbb{C}$



$\arg\left(\frac{z}{\omega}\right)$

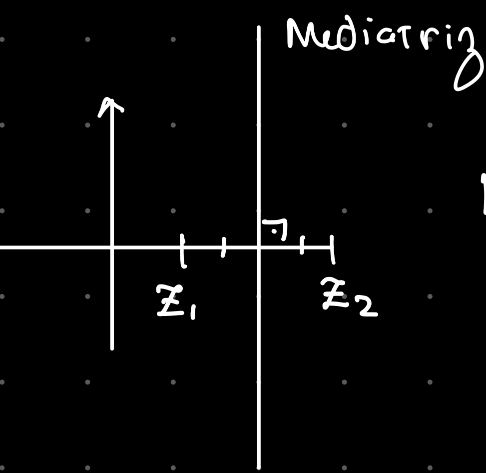
$|z+\omega|^2 = |z|^2 + |\omega|^2 + 2|z||\omega| \cdot \cos\theta$

Desigualdade Triangular

$$\big|\,|z|-|w|\,\big| \leqslant |z+w| \leqslant |z|+|w|$$

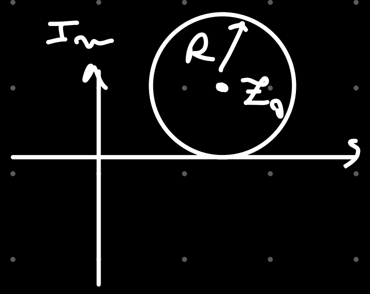
Lugar Geométrico

1) Mediatriz



LG Mediatriz: $|z - z_1| = |z - z_2|$

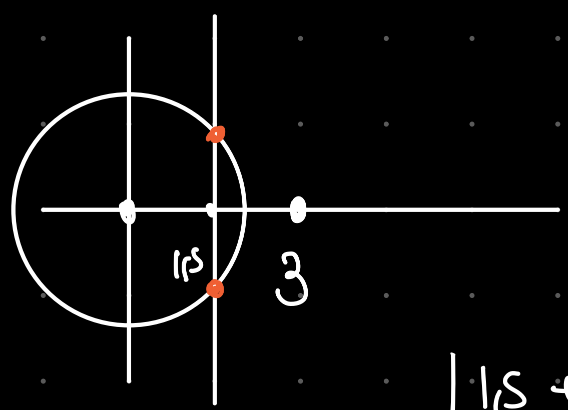
2) Circunferência



LG circ : $|z - z_0| = R$

com $R > 0$

Exercício Solucione: $|z| = \dfrac{4}{|z|} = |z-3|$



$$\begin{cases} |z|^2 = 4 \\ |z-3| = |z| \end{cases}$$

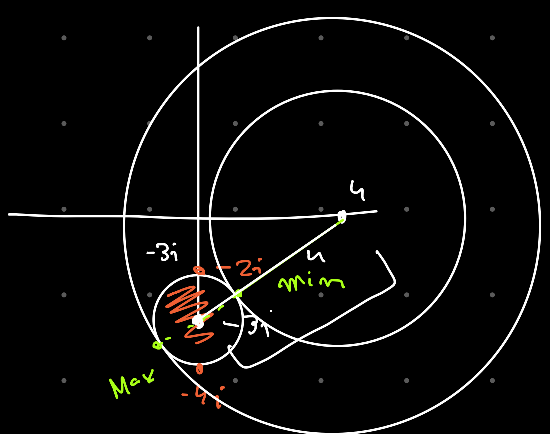
$$|1{,}5 + bi| = 2$$

$$\sqrt{\left(\tfrac{3}{2}\right)^2 + b^2} = 2$$

$$\frac{9}{4} + b^2 = 4$$

$$z = \frac{3}{2} \pm \frac{\sqrt{7}}{2} i$$

Exercício Determine $|z-4|_{max}$ e $|z-4|_{min}$ dado $|z+3i| \le 1$



4, 6

Exercício Determine z tal que

$$\frac{\bar{z}}{z-2i} + \frac{2z}{\bar{z}+2i} = 3 \quad e \quad 0 < |z-2i| \leq 1$$

$$\omega + 2\bar{\omega} = 3$$

$$x+yi+2x-2yi = 3 \rightarrow 3x - yi = 3 \quad \begin{cases} x = 1 \\ y = 0 \end{cases}$$

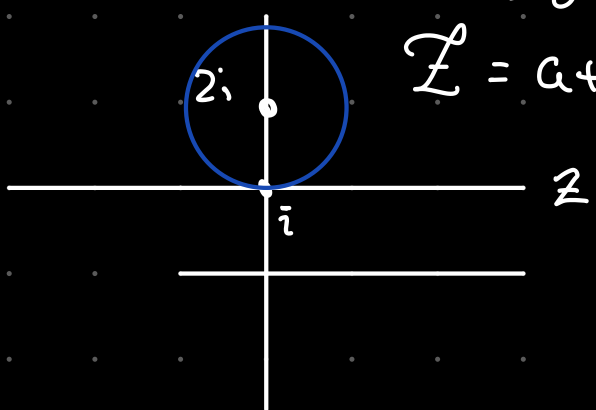
$$\omega = 1$$

$$\frac{\bar{z}}{z-2i} = 1 \qquad \bar{z} = z - 2i \rightarrow$$

$$\cancel{a} - bi = \cancel{a} + bi - 2i$$

$$2bi = 2i \Rightarrow b = 1$$

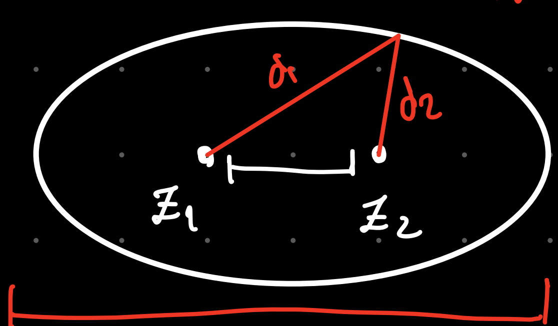
$$z = a + i, \quad a \in \mathbb{R}$$



$$z = i$$

3) Elipse

$$d_1 + d_2 = 2a$$



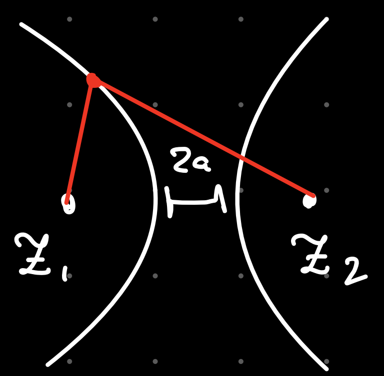
$$|z - z_1| + |z - z_2| = 2a$$

em que

$$2a > |z_1 - z_2|$$

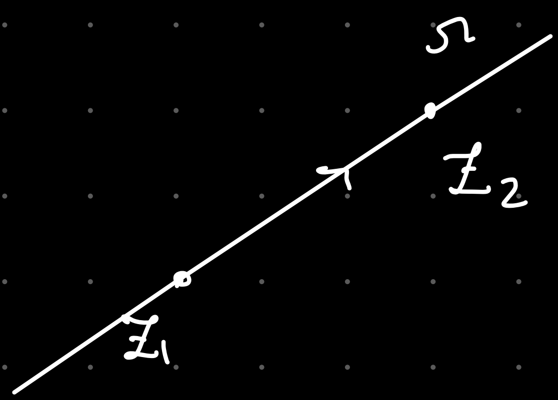
$2a$

4) Hipérbole



$$||z-z_1| - |z-z_2|| = 2a$$

5) Eq de Reta



$$r: z_1 + \lambda(z_2 - z_1)$$

$$r: (1-\lambda)z_1 + \lambda z_2 = z$$

$$\lambda \in \mathbb{R}$$

Ex IME 1ª Fase Seja $\alpha \in \mathbb{R}$ e $z_1, z_2, z_3 \in \mathbb{C}$ $z_1 \neq z_2$ tais que $|z_1| = |z_2| = |z_3| = 4$

O menor valor de

$$|\alpha z_1 - (\alpha - 1)z_2 - z_3| \text{ é ?}$$

a) $\frac{1}{4}|z_1 + z_2|$

$$|(1-\alpha)z_2 + \alpha z_1 - z_3|$$

b) $\frac{1}{4}|z_1-z_2|$

c) $\frac{1}{8}|z_3-z_1||z_3-z_2|$ h

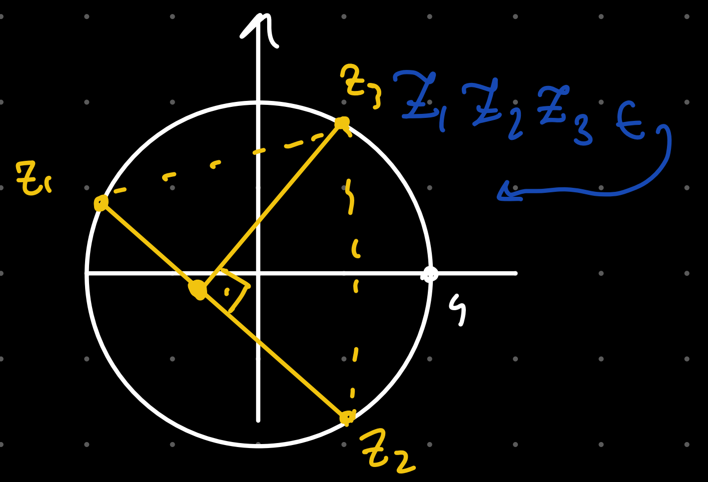
d) $\frac{1}{4}|z_1-z_2-z_3|$

e) $|z_3|$

Eq de reta

distancia do ponto à reta

é um $z \in$ Reta para z_1, z_2



$$A_\Delta = |z_1-z_2| \cdot \frac{h}{2} = \frac{P}{R} \to 4$$

$$\frac{h\,|z_1-z_2|}{2} = \frac{|z_1-z_3||z_2-z_3||z_1-z_2|}{8}$$

$$h = \frac{1}{4}|z_1-z_3||z_2-z_3|$$

Números Binômiais

número binomial representa $P_n^{\alpha, n-\alpha}$

$$\binom{n}{p} = \frac{n!}{p!\,(n-p)!}$$

Triangulo de Pascal

$$\begin{matrix} \binom{0}{0} \\ \binom{1}{0}\binom{1}{1} \\ \binom{2}{0}\binom{2}{1}\binom{2}{2} \\ \binom{3}{0}\binom{3}{1}\binom{3}{2}\binom{3}{3} \\ \binom{4}{0}\binom{4}{1}\binom{4}{2}\binom{4}{3}\binom{4}{4} \end{matrix} = \begin{matrix} 1 \\ 1 \quad 1 \\ 1 \quad 2 \quad 1 \\ 1 \quad 3 \quad 3 \quad 1 \\ 1 \quad 4 \quad 6 \quad 4 \quad 1 \end{matrix}$$

Propriedades Binomiais

P1) Binomiais Complementares

escolher uma coisa é abrir mão das outras

$$\binom{n}{p} = \binom{n}{n-p} \quad \therefore \quad \binom{18}{7} = \binom{18}{11}$$

Ex.: $\binom{16}{10} = \binom{16}{x} \rightarrow x = 6$ ou $x = 10$

P2) Relação de Stiffel

1

1 + 1

1 + 2   1

1   3 + 3 + 1

1   4   6   4   1

• $\binom{n}{p} + \binom{n}{p+1} = \binom{n+1}{p+1}$

Demo: $\frac{n!}{(n-p)!\,p!} + \frac{n!}{(n-p+1)!\,(p+1)!}$

$$= \frac{n!}{(n-p)!\,p!} + \frac{n!}{(n-p)!\,(n-p+1)\,p!\,(p+1)}$$

$$= \frac{n!}{(n-p+1)!\,(p+1)!}\Big((n-p+1)(p+1)+1\Big) = \frac{n!}{(n-p+1)!\,(p+1)!}\Big(np+n-p^2-\cancel{p}+\cancel{p}+1+1\Big)$$

$$= \frac{n!}{(n-p+1)!\,(p+1)!}\Big(n(p+1)+1-p^2\Big) = \frac{(n+1)!}{(n-p+1)!\,(p+1)!}$$

Argumento Combinatório:

escolha p+1 animais em um grupo de n cães e 1 gato

$$\binom{n+1}{p+1} = \underbrace{\binom{n}{p}}_{\text{com o gato}} + \underbrace{\binom{n}{p+1}}_{\text{sem o gato}}$$

P3) Teorema das Linhas

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n$$

Conforme o número de partículas aumenta (n) o número de microestados aumenta 2^n

Exemplo Calcule $\binom{100}{1} + 2\binom{100}{2} + \cdots 100\binom{100}{100}$

$$\therefore \sum_{n=1}^{100} n\binom{100}{n} \qquad TG.: \ n\binom{100}{n}$$

$$n\left(\frac{100!}{(100-n)!\,n!}\right) = \frac{100!}{(100-n)!\,(n-1)!}$$

$$100\left(\frac{99!}{(99-n+1)!\,(n-1)!}\right) = 100\binom{99}{n-1}$$

$$100\sum\binom{99}{n-1} = \underline{100\cdot 2^{99}}$$

P4) Relação de Euler

$$\binom{m}{0}\binom{n}{p} + \binom{m}{1}\binom{n}{p-1} + \cdots + \binom{m}{p}\binom{n}{0} = \binom{m+n}{p}$$

Para um caso particular $m = n = p$

$$\underbrace{\binom{n}{0}\binom{n}{n}} + \binom{n}{1}\binom{n}{n-1} + \cdots + \binom{n}{n}\binom{n}{0} = \binom{n+n}{n}$$

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \binom{n}{2}^2 + \cdots \binom{n}{n}^2 = \binom{2n}{n}$$

Ex 2 Gauss pode dar passos nas direções N, S, L, O de quantos modos Gauss pode voltar ao ponto original depois de 2024 passos

seja a o número de passos na direção N
seja b o número de passos na direção S
seja c na leste
seja d na oeste

$$a + b + c + d = 2024$$
$$a = b \quad e \quad c = d$$
$$2a + 2c = 2024$$
$$a + c = 1012$$
.... / ...
$$P_{1013}^{1012,1} = \frac{1013!}{1012!\,1!} = 1013$$

Sol JM

Passos
$$\begin{cases} N: a \\ S: a \\ L: b \\ O: b \end{cases}$$

$$2a + 2b = 2024$$

$$a + b = 1012$$

dado a passos North

$$\underbrace{NNNN}_{a}\ldots\underbrace{SSSSS}_{a}\ldots\underbrace{LLLLLLL}_{b}\ldots\underbrace{OOOOOO}_{b}$$

Total é o número de anagramas de

$$\text{Total} = \frac{(2a+2b)!}{a!\,a!\,b!\,b!} = \frac{2024!}{(a!)^2[(1012-a)!]^2}$$

$$\text{Total} = 2024!\left(\frac{1}{(a!)^2[(1012-a)!]^2}\right)$$

$$\text{Total} = \frac{2024!}{(1012!)^2}\left[\frac{1012!}{a!\,(1012-a)!}\right]^2$$

$$\text{Total} = \binom{2024}{1012}\binom{1012}{a}^2$$

Como a pode ser $0, 1, 2, 3, \ldots, 1012$

$$\text{Total} = \binom{2024}{1012}\left[\binom{1012}{0}^2 + \binom{1012}{1}^2 + \binom{1012}{2}^2 + \binom{1012}{3}^2 + \ldots + \binom{1012}{1012}^2\right]$$

$$\text{Total} = \binom{2024}{1012}\binom{2024}{1012} = \binom{2024}{1012}^2$$

PS) Teorema das Colunas

$$\begin{array}{ccccc}
\binom{0}{0} & & & & \\
\binom{1}{0} & \binom{1}{1} & & & \\
\binom{2}{0} & \binom{2}{1} & \binom{2}{2} & & \\
\binom{3}{0} & \binom{3}{1} & \binom{3}{2} & \binom{3}{3} & \\
\binom{4}{0} & \binom{4}{1} & \binom{4}{2} & \binom{4}{3} & \binom{4}{4}
\end{array}
\qquad
\begin{array}{ccccc}
1 & & & & \\
1 & 1 & & & \\
1 & 2 & 1 & & \\
1 & 3 & 3 & 1 & \\
1 & 4 & 6 & 4 & 1
\end{array}$$

$$\binom{n}{n}+\binom{n+1}{n}+\binom{n+2}{n}\cdots\binom{n+p}{n}=\binom{n+p+1}{n+1}$$

Demo:

$$\binom{n}{n}=\binom{n+1}{n+1}$$

$$\binom{n+1}{n+1}+\binom{n+1}{n}=\binom{n+2}{n+1}$$

$$\binom{n+2}{n+1}+\binom{n+2}{n}=\binom{n+3}{n+1}$$

...

P6) Teorema das Diagonais

$$\begin{array}{ccccc}
\binom{0}{0} & + & & & \\
\binom{1}{0} & \binom{1}{1} & + & & \\
\binom{2}{0} & \binom{2}{1} & \binom{2}{2} & + & \\
\binom{3}{0} & \binom{3}{1} & \binom{3}{2} & \binom{3}{3} & \\
\binom{4}{0} & \binom{4}{1} & \binom{4}{2} & \binom{4}{3} & \binom{4}{4}
\end{array}$$

$$\begin{array}{ccccc}
1 & & & & \\
1 & 1 & & & \\
1 & 2 & 1 & & \\
1 & 3 & 3 & 1 & \\
1 & 4 & 6 & 4 & 1
\end{array}$$

$$\binom{n}{0}+\binom{n+1}{1}+\binom{n+2}{2}+\cdots+\binom{n+p}{p}=\binom{n+p+1}{p}$$

$$\binom{n}{n}+\binom{n+1}{n-1}+\binom{n+2}{n-2}+\cdots+\binom{n+p}{n-p}=\binom{n+p+1}{n+1}$$

é complementar do das colunas

P7) Relação de Fermat

$$\frac{n-p}{p+1}\binom{n}{p} = \binom{n}{p+1}$$

Ex ITA 2014 Calcule

$$\binom{n}{0} + \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} + \dots + \frac{1}{n+1}\binom{n}{n}$$

T.G. $\frac{1}{p+1}\binom{n}{p} = \frac{1}{p+1}\frac{n!}{(n-p)!\,p!}$

$$= \frac{(n+1)!}{(p+1)!\,(n-p)!}\cdot\frac{1}{n+1}$$

$$= \frac{1}{n+1}\sum\binom{n+1}{p}$$

Binômio de Newton

considere as potencias dos Binômios de Newton

$(x+a)^0 = 1$

$(x+a)^1 = 1x + 1a$

$(x+a)^2 = 1x^2 + 2xa + 1a^2$

$(x+a)^3 = 1x^3 + 3x^2a + 3xa^2 + 1a^3$

$(x+a)^4 = 1x^4 + 4x^3a + 6x^2a^2 + 4xa^3 + 1a^4$

generalizando

$$(x+a)^n = \binom{n}{0}x^n a^0 + \binom{n}{1}x^{n-1}a^1 + \binom{n}{2}x^{n-2}a^2 + \dots + \binom{n}{p}x^{n-p}a^p + \dots + \binom{n}{n}x^0 a^n$$

$$(x+a)^n = \sum_{p=0}^{n} \binom{n}{p} x^{n-p} a^p$$

assim $(x+a)^n$ tem $n+1$ parcelas.

a Fórmula do Termo geral é

$$T_{p+1} = \binom{n}{p} x^{n-p} a^p$$

começa no 0

Exercício: Para o desenvolvimento de $(3x-2)^{10}$ Calcule

a) a parte numérica de x^3

$$\binom{n}{p}(3x)^{n-p}(-2)^p \rightarrow \begin{matrix} n=10 \\ p=7 \end{matrix} \rightarrow \binom{10}{7} 3^3 (x)^3 (-2)^7$$

$$\text{Resposta} = -\binom{10}{7} 27 \cdot 2^7$$

b) Σ dos Coeficientes

Σ coeficientes $x \rightarrow 1$ $\quad (3-2)^{10} = 1$

Exemplo 2: Determine o coef de x^5

No produto $(1+x)^4(1+x^2)^5$

$$\left[\binom{n}{p}x^{n-p}(1)^p\right]\left[\binom{\varphi}{\pi}(x^2)^{\varphi-\pi}(1)^\pi\right] \qquad \begin{matrix} n=4 \\ \varphi=5 \end{matrix}$$

$$n-p+2\varphi-2\pi=4 \qquad \binom{n}{p}\binom{\varphi}{\pi}$$

$$14-p-\cancel{4}\pi=4$$

$$p+2\pi=10 \qquad \binom{n}{10-2\pi}\binom{\varphi}{\pi}$$

Ex.: Nº de Termos Inteiros na Expansão do

$$\left(\sqrt{5}+\sqrt[4]{11}\right)^{124}$$

$$\binom{n}{p}\left(\sqrt{5}\right)^{n-p}\left(\sqrt[4]{11}\right)^{p} \qquad \text{Inteiro geral} \qquad n=124$$

$2 \mid 124-p$

$4 \mid p$; 31 Termos

$$\begin{array}{r|l} 124 & 4 \\ \underline{12} & \overline{31} \\ 04 & \end{array}$$

4 8 12 16 20 24

32 28 32 36 40 ...

Ex.: Coef do x^{20} em $(1+3x+3x^2+x^3)^{20}$

$$\left((1+x)^3\right)^{20} \qquad (1+x)^{60}$$

$$60-p=20$$

$$\binom{60}{} x^{60-p}\,1^{p} \qquad p=40$$

$\binom{60}{30}$ $\binom{60}{20}$

Ex 5.: Calcule coef x^4 em

$(1+x)^{4}+(1+x)^{12}+(1+x)^{13}+\dots+(1+x)^{20}$

$\binom{11}{p}x^{11-p}$ $\binom{11}{7}+\binom{12}{8}+\binom{13}{9}\dots+\binom{20}{16}$

$11-p=4 \to p=7$
$12-p=4$ $p=8$
$13-p=4$ $p=9$
10
11
12
13
16

Diagonal

$$\begin{array}{ccc} n & & n+p \\ \binom{4}{0} & \dots & \binom{20}{16_p} \end{array}$$

$\binom{21}{16}-\binom{11}{6}$ $-\binom{4}{0}\dots\binom{10}{6}$

Ex 6.: Quais 2 termos consecutivos do $(3+2x)^{74}$ possuem coef iguais

$\binom{74}{p}(3)^{74-p}(2)^{p}(x)^{p}=\binom{74}{c}(3)^{74-c}(2)^{c}(x)^{c}$

$\binom{74}{c}(3)^{74-c}(2)^{c}=\binom{74}{p}(3)^{74-p}(2)^{p}$

Exercício IME 2021 seja polinômio

$1+y+y^2 \dots y^{20}$ que pode ser escrito

com $\alpha_0+\alpha_1 x+\alpha_2 \dots \alpha_{20} x^{20}$ em que $x = y+1$

Calcule α_3 $\qquad \alpha_3 x^3 \qquad x-1=y$

$$1 \qquad (y+1)^3 \qquad (x-1)^3 = y^3$$

$$y^3+y^2+y+1 \qquad x^3-x^2+x-1=y^3$$

Equação Multinomial (Leibniz)

Dado $n \in \mathbb{N}$ Temos:

$$(x_1+x_2+x_3+\dots+x_p)^n = \sum_{\alpha_1,\alpha_2,\dots\alpha_p}^{\alpha_1+\alpha_2\dots=n} \frac{n!}{\alpha_1!\alpha_2!\dots\alpha_p!} x_1^{\alpha_1} x_2^{\alpha_2} \dots x_p^{\alpha_p}$$

onde $\sum \alpha_i = n \quad \alpha_i \in \mathbb{N}$

Exemplo qual coef de $x^2 y z^2$ no desenvolvimento de $(x-2y+z)^5$

$$(x-2y+z)^5 = \sum_{a,b,c}^{a+b+c=5} \frac{5!}{a!b!c!} x^a (-2y)^b z^c$$

assim para $a=2$, $b=1$ e $c=2$

$$\frac{5!}{2!\,1!\,2!} x^2 (-2y)^1 z^2 = -60x^2 y z^2$$

Ex$_2$ Determinar coef de x^4 em $(x^2-x+2)^6$

$$\frac{6!}{a!\,b!\,c!}(x^2)^a(-x)^b(2)^c$$

$$2a+b+c = 6$$

and $c = 2$

$$\begin{cases} a=0,\ b=4 \\ a=1,\ b=2 \\ a=2;\ b=0 \end{cases}$$

$$\frac{6!}{0!\,4!\,2!}2^2(-1)^4 + \frac{6!}{1!\,2!\,3!}2^3(-1)^2 + \frac{6!}{2!\,0!\,4!}(2)^4(-1)^0 = 780$$

Ex$_3$ Quantos termos possui o desenvolvimento

$(x+y+z+w)^{10}$?

$$\sum_{a,b,c,d\in\mathbb{N}}^{a+b+c+d=10} \frac{10!}{a!\,b!\,c!\,d!}(x)^a(y)^b(z)^c(w)^d$$

de quantas formas $a+b+c+d=10$ → Diofantina)

$\longrightarrow$ ●●●●●●●●●●///

$$P_{13}^{3,10} = \frac{13!}{10!\,3!} = \frac{13.12.11}{3.2.1} = 13\cdot 22 = 286$$

Somatórios Binomiais

Considere o desenvolvimento:

$$(1+x)^n = \sum_{i=0}^{n} \binom{n}{i} x^i$$

$x \to 1$ $\quad (1+1)^n = 2^n = \sum_{i=0}^{n} \binom{n}{i} = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} \cdots$

assim $\quad 2^n = \binom{n}{0} + \binom{n}{1} + \cdots \binom{n}{m}$ $\quad$ Soma da linha (I)

$x \to -1$

$$(1-1)^n = 0 = \sum_{i=0}^{n} \binom{n}{i} (-1)^n = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} \cdots + \binom{n}{n}(-1)^n$$

soma alternando sinal $\quad$ (II)

$\dfrac{(I)+(II)}{2}$: $\quad \dfrac{1}{2} = \binom{n}{0} + \binom{n}{2} + \binom{n}{4} \cdots$

$$\frac{1}{2} = \sum_{i=0}^{2i \leq n} \binom{n}{2i}$$

Soma dos pares (III)

$\dfrac{(I)-(II)}{2}$: $\quad \dfrac{1}{2} = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} \cdots$

$$\frac{1}{2} = \sum_{i=0}^{2i-1 \leq n} \binom{n}{2i-1}$$

(IV)

e se $x \to i$

$$(1+i)^n = \binom{n}{0} i^0 + \binom{n}{1} i + \binom{n}{2} i^2 + \binom{n}{3} i^3 + \binom{n}{4} i^4 + \binom{n}{5} i^5 + \cdots$$

$$= \binom{n}{0} + \binom{n}{1} i - \binom{n}{2} - \binom{n}{3} i + \binom{n}{4} + \binom{n}{5} i + \dots$$

$$= \left[\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \binom{n}{6}\right] + i\left[\binom{n}{1} - \binom{n}{3} + \binom{n}{5} - \binom{n}{7} \dots\right]$$

Sendo $1+i = \sqrt{2}\,\mathrm{cis}\frac{\pi}{4} \to (i+1)^n = \sqrt{2}^n\,\mathrm{cis}\frac{n\pi}{4}$

$$\sqrt{2}^n\left[\cos\frac{n\pi}{4} + i\sin\frac{\pi}{4}\right]$$

$$\sqrt{2}^n \cos\frac{n\pi}{4} = \left[\binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots\right] \quad (\mathrm{V})$$

$$\sqrt{2}^n \sin\frac{n\pi}{4} = \left[\binom{n}{1} - \binom{n}{3} + \binom{n}{5} - \binom{n}{7} + \dots\right] \quad (\mathrm{VI})$$

$$\frac{(\mathrm{III} + \mathrm{V})}{2} : \binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \binom{n}{6} + \dots \Big] =$$

$$+ \binom{n}{0} - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots \Big] \frac{1}{2}$$

$$= \frac{2^{n-1} + \sqrt{2}^n \cos\left(\frac{n\pi}{4}\right)}{2}$$